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14 September 1990
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15 April 1971

MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADAR RT-860/VPS-2: 10548089

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and Is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Receiver-Transmitter, Radar, RT-860/VPS-2 , hereinafter referred to as a receiver-transmitter, which is part of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel, Manufacture and Inspection, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent tdata which may be of use in improving this document should be be addressed to: Commander, U.S.Army ARDEC, ATTN:SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
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MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-M-48428	Microwave Chassis Assy: 10549047
MIL-T-50320	Transmit Gate Assy: 10549202
MIL-P-50321	Klystron Power Supply Assy: 9360497
MIL-P-50323	Klystron Beam Power Supply Assy: 11834450
MIL-P-50324	Pulse Shaper Assy: 10548801
MIL-R-50651	Radar Set AN/VPS-2: 10549058

STANDARDS

MILITARY

MIL-STD-109	**Quality Assurance Terms and Definitions**
MIL-STD-810	**Environmental Test Methods and Engineering Guidelines**
MIL-STD-2000	**Standard Requirements for Soldered** Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER **(ARDEC)**

10548089	Receiver-Transmitter, Radar RT-860/VPS-2
10548242	Relay Time Delay, SPNO

PACKAGING DATA SHEET

PS10548089	Packaging of Receiver-Transmitter, Radar RT-860/VPS-2

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548089, all associated drawings, and with all requirements specified in applicable specifications and standards. The subassemblies of the receiver-transmitter shall **have** been inspected and accepted in accordance with the drawing prior to assembly in the receiver-transmitter. The following subassemblies are covered by the indicated specifications:

Unit No.	Subassembly	Specification
2A1	Microwave Chassis Assy: 10549047	MIL-M-48428
2A2	Klystron Power Supply Assy: 9342815	MIL-P-50321
2A5	Transmit Gate Assy: 10549202	MIL-T-50320
2A6	Klystron Beam Power Supply Assy: 11834450	MIL-P-50323
2A7	Pulse Shaper Assy: 10548801	MIL-P-50324

Where applicable, the receiver-transmitter shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. Workmanship

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3.3 Environmental service conditions.

3.3.1 Vibration. The receiver-transmitter shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to **15** hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ±10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ±10%.

3.3.2 Shock. The receiver-transmitter **shall** not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ±10%, 11 milliseconds ±10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 OPerating. The receiver-transmitter shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The receiver-transmitter shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The receiver-transmitter shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The receiver-transmitter **shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned** to room ambient.

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3.3.5 Humidity (operating) . The receiver-transmitter shall meet the requirements of 3.4 and 3.5 during exposure to an atmosphere with a relative humidity In excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the receiver-transmitter.

3.5 Performance. The unit under test and all test **equipment shall be** allowed 15 minutes warm-up time prior to testing.

3.5.1 Interlock switch. When the receiver-transmitter cover is open, an open circuit shall exist between 2JI-F and 2J1-H. When the receiver-transmitter cover is closed, continuity shall exist between 2JI-F and 2J1-H. A manual override feature shall be included in the interlock switch to provide continuity between 2J1-F and 2J1-H when the receiver-transmitter cover is open and the interlock switch plunger is In the manual override position.

3.5.2. Clutter lock-on switch. The clutter lock-on switch shall provide continuity between 2J1-b and 2J1-Z when in the normal position. There shall not be continuity between 2J1-b and 2TB1-1. Alternatively, when this clutter lock-on switch is in the test position, there shall be continuity between 2J1-b and 2TB-1. There shall be no continuity between 2J1-b and 2J1-Z.

3.5.3 Overload reset switch. When the overload reset switch is operated to either of its reset positions from the spring-loaded center off position, the switch shall provide continuity between 2TB1-1 and 2P18-2.

3.5.4 Indicator lamps.

3.5.4.1 Modulator overload indicator. The modulator overload lamp, 2DSl, shall illuminate when 28 volts dc (Vdc) is applied between 2P18-3 and 2TB1-2.

3.5.4.2 High voltage overload indicator. The high voltage overload lamp, 2DS2, shall illuminate when 28 **Vdc is** applied between 2TB1-2 and 2P18-1.

3.5.4.3 Power indicator lamp. When 24 Vdc is applied between 2J1-G and 2J1-Y and the door interlock switch is closed, the power indicator lamp, 2DS-3 shall illuminate.

3.5.5 Standby operation.

3.5.5.1 Blower operation. The blower, 2B1, shall operate and provide an intake airflow.

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3.5.5.2 Gated stalo output. A gated 60 MHz stalo signal **shall be** present at 2J2-G and shall exhibit the following parameters:

Amplitude: 0.1 V peak-to-peak minimum

Width : (50% points) 280 nanoseconds (ns)

3.5.5.3 Pulse shaper output. The 2A7 pulse shaper output shall appear at 2P23 with the following parameters:

PRF: 20 kHz ± 0.1 kHz

Amplitude: +15.o v peak minimum

Width : *Minimum . 0.80 US @ 60°F tO 90°F
Maximum - 1.30us @ 60 F to 90 F

*Minimum . **0.60** us @ -40°F

3.5.5.4 Overall gain. The overall gain of the receiver-transmitter, measured from 2J3 to 2J5-D with 0.0 Vdc at 2J5-E, shall be 17 decibels (dB) minimum, 22 dB maximum.

3.5.5.5 AGC action. The gain of the receiver-transmitter shall be reduced by a minimum of 20 dB when the dc voltage at 2J5-E is changed from 0.0 Vdc to +6.0 Vdc. The gain versus AGC voltage curve shall be monotonic.

3.5.5.6 NOise figure. The noise figure for the receiver-transmitter shall be 12 dB maximum on all six channels.

3.5.5.7 Switch drive Pulse. The switch drive pulse at 2A1A7J1 shall exhibit the following parameters.

Negative amplitude:	22 Vdc minimum
Positive reference:	+0.7 Vdc minimum, +1.1 Vdc maximum
Width: (50% points)	2.48 us minimum, **2.62 us maximum**

3.5.5.8 **Diode switch isolation. When the dc level at 2A1A7J1 is changed from +1.0 Vdc to -33.0 Vdc, the insertion loss of the diode switch shall increase a minimum of 45 dB.**

3.5.6 **Radiate** mode.

3.5.6.1 Average output power. The average output power, measured at 2J3, shall be 4.26 watts (W), minimum.

3.5.6.2 Transmit pulse timing. The transmit pulse timing shall be such that **the** delay between the Klystron modulator pulse **edge at the start** of the flat and the center of the gated 60 MHZ RF (COS^{9}) pulse, shall **be 0.3 us minimum, 0.32 us** maximum.

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3.5.6.3 Transmit pulse envelope. The transmit pulse envelope, when detected and observed on an oscilloscope, shall have a cosine squared (COS^2) characteristic shape and shall be 280 ns wide (set) at the 50 percent points.

3.5.6.4 Transmit frequency. The receiver-transmitter unit shall be capable of operation on six discrete frequencies by selecting one of six oscillator plug-in modules. The frequency of the transmitted signal shall be 9205 MHz ±10 MHz using Charnel 1 oscillator module. Using the Channel 1 measurement as a reference, the charnel 2 through 6 output frequency shall be as follows:

Chan. 2	Reference +8 ±2 MHz
Chan. 3	Reference +16 ±2 MHz
Chan. 4	Reference +24 ±2 MHz
Chan. 5	Reference +32 ±2 MHZ
Chan. 6	Reference +40 ±2 MHz

3.5.6.5 Transmit indicator threshold. The receiver-transmitter shall provide +27.0 ± 1.0 Vdc at 0.25 amps between pins 2J7-B(+) and A(-) when the average output power measured at 2J3 exceeds 1.9 watts(w). When the average output power measured at 2J3 is less than 1.7 watts(W) the voltage between pins 2J7-B(+) and A(-) shall be less than +2.0 Vdc.

3.5.7 Subclutter visibility (SCV) test. The receiver-transmitter after having met the requirements of 3.5.1 through 3.5.6 shall meet the subclutter visibility (SCV) requirements as specified in MIL-R-50651 when substituted into a Radar Set AN/vPs-2. Radar Set AN/vPs-2 and facilities shall be as specified:

a. * SCV without AGC -43db minimum at minimum discernible signal.

b. * SCV with AGC -41 ± ldb **at** minimum discernible signal.

c. * Hi-Level (-15dbm clutter) with AGC -34db minimum, -42db **maximum.**

* Lock threshold 8db above system standby noise level.

3.6 First article. When specified in the contract **or** purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. The receiver-transmitter shall be soldered per general requirements and Tasks A, C, and D of MIL-STD-2000.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-sTD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items must meet **all** requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become **a** part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, **nor does it commit the Government to accept defective material.**

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the **quality** assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 **First article inspection.**

4.3.1 First article submission. The contractor shall submit a first ***article*** sample as designated **by the** contracting officer consisting of three receiver-transmitters for Inspection in accordance with 4.3.2.

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4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional Test) and Table II (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. "

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components. assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

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4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling One receiver/transmitter out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table I and shall then **meet the tests** in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104 . Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107 . Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last teSt as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the 'Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE III. Test equipment list

ITEM NUMBER	DESCRIPTION
1. X-band rf load	Band width: 9150 to 9250 MHz minimum. Average power rating: 10 W minimum. VSWR : **1.0:1.1 maximum.**
2. X-band detector	Bandwidth: 9150 to 9250 MHz minimum. Average power rating: 10 mw minimum VSWR : 1.0:1.8 maximum with 100-ohm load and 0.5 mA crystal current.
3. X-band coupler (2 required)	Bandwith: 9150 to 9250 MHz minimum. Peak power rating: 2000 W minimum. Coupling: 40 dB ±0.9 dB. Mainline VSWR: 1.0:1.1 maximum. Secondary line VSWR: 1.0:1.2 maximum.

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TABLE III . Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
4*	X-band signal generator	Frequency range: 9150 MHz to 9250 MHz minimum. Frequency stability: with temperature- less than 0.006%/ deg.C change in ambient temperature; with line voltage - less than 0.02% change for line voltage variation of +10%. Residual FM: less than 15 KHz **peak.** **Output range: 0.224 V to 0.1 uV** rms into a **50-ohm load.** **Output calibration: 0 dBm to -127 dBm.** **Source impedance: 50 ohms nominal.**
5.	X-band frequency meter	Type: direct reading resonant cavity. Frequency range: 9150 to 9250 MHZ minimum. Overall accuracy: 0.08%.
6.	Thermistor mount	Frequency range: 9150 to 9250 MHZ minimum. Maximum average input power: 10 mW minimum.
7.	Power meter	Frequency range: 9150 to 9250 MHZ minimum. Power range: 1 mW full scale. Accuracy: ±1% of full scale. Type: Thermistor sensing element.
8.	I-f amplifier	Frequency: 60MHz. 3 dB bandwith: 10 MHZ minimum. Gain: 20 dB.. Maximum Input: not less than 0 dBm with a maximum distortion of 5%
9.	Noise figure meter	Type : automatic readout with argon tube waveguide noise source and 60 MHz integral i-f amplifier. Scale: not less than 20 dB full scale; 12 dB calibration point shall be not less than half scale.

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TABLE 111. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
10.	Waveguide noise source	Type: argon-filled gas-discharge tube in standard waveguide housing. Output connections: UG-39/U. Excess noise rating: 15.75 ±0.5 dB.
110	Oscilloscope	Vertical display: dual trace, dc to 75 MHz (approximately 3 dB down) at 10 mV/cm minimum. Horizontal sweep: calibrated time base 50 ns/cm to 1 s/cm minimum. Delayed sweep: 2 us to 50 us, Continuously variable, minimum.
12.	Power supply	Type: laboratory supply. Minimum requirements: 0 to 32 Vdc. Load regulation: 0.02%. Load current capability: 15 A. Ripple and noise: 1 mV rms or 5 mv peak-to-peak maximum from dc to 20 MHz.
13.	Power supply	Type: laboratory supply. Minimum requirements: 0 to 30 Vdc. Load regulation: 0.02%. Load current capability: 5 A. Ripple and noise: 1 mV rms or 5 mV peak-to-peak maximum from dc to 20 MHz.
14.	Power supply	Type: laboratory supply, independent dual voltage source. Minimum requirements: 0 to 20 Vdc. Load regulation: 0.02%. Load current capability: 100 mA. Ripple and noise: 1 mV rms or 5 mv peak-to-peak maximum from dc to 20 MHz.
15.	Power supply	Type: laboratory supply. Minimum requirements: 0 to 40 Vdc. Load regulation: 0.02%. Load current capability: 100 mA. Ripple and noise: 1 mV rms or 5 mv peak-to-peak maximum from dc to 20 MHz.

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TABLE III. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
16.	Rf voltmeter	**Ranges: 3 mv to 300 mV** in four ranges. Frequency range: 1 MHz to 100 MHz minimum. Accuracy: ±3% of full scale.
17.	Digital voltmeter	Display: four digit readout. Accuracy: 0.01% **Range: 100 Vdc.**
18.	Pulse generator	Pulse amplitude: **+15 V** peak. Pulse width: 0.4 us. Rise time: 100 ns **maximum.**
19.	Text fixture	**See Figure 1 and Table IV.**
20.	Rf attenuator	Frequency range: dc to 100 MHz. Attenuation: **0-30** dB in 1 dB steps.

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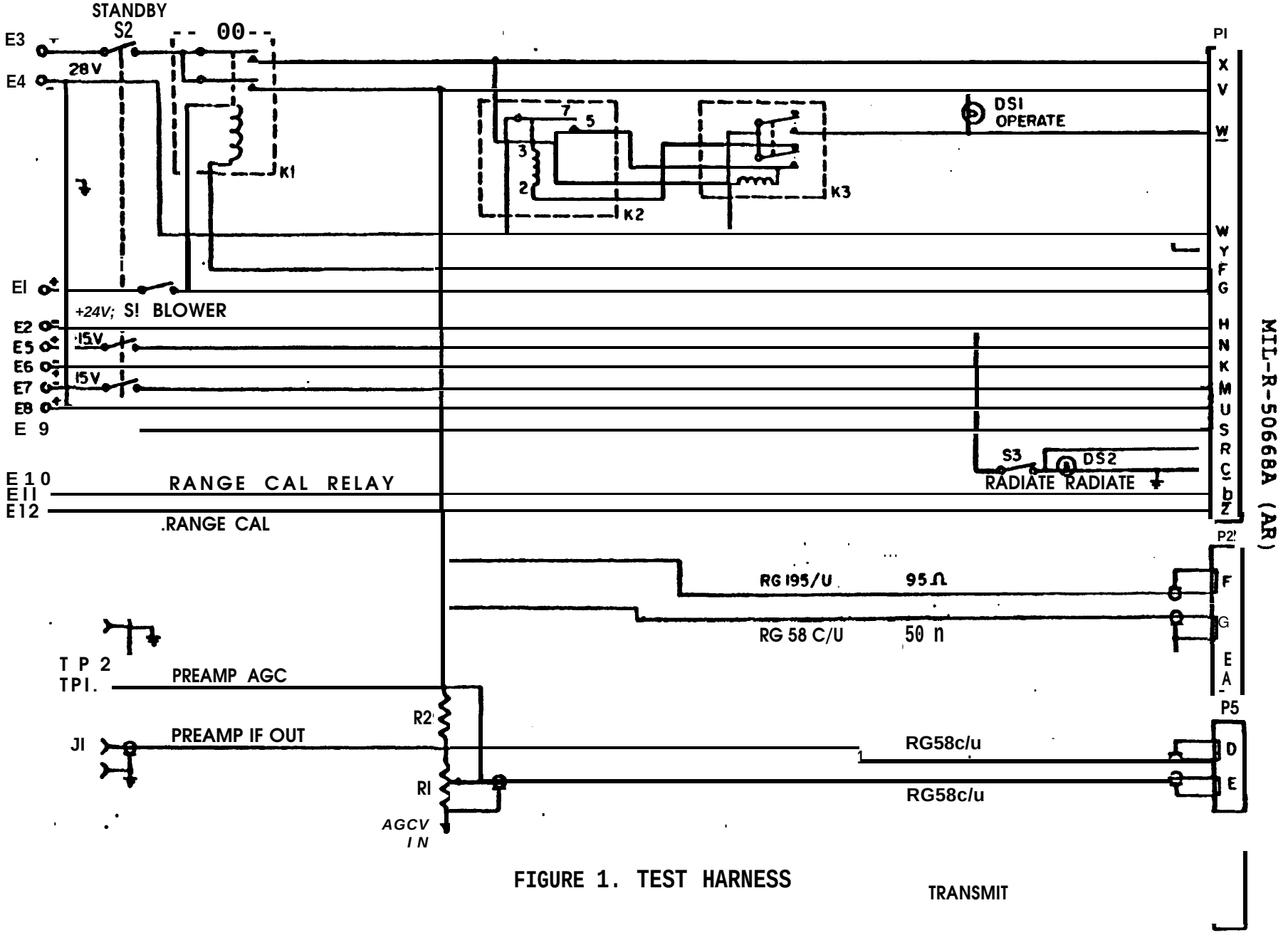


FIGURE 1. TEST HARNESS

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TABLE IV. Test harness **parts list**

DESIGNATION	PART	DESCRIPTION
DS1, DS2, DS3	Lamp	No. 313
El through E12	Binding post	H. H. Smith, 257
Jl, J2, J3	Connector, BNC	UG-625B/U
Kl, K3	Relay	Potter and Brumfield, KA1lDG
K2	Relay, time delay	10548242
P1	Connector	Bendix, PT06A20-26S(SR), or equivalent
P2	Connector	Bendix, PT06SE22-78S(SR), or **equivalent.**
P5	Connector	**Bendix, PT06SE22-78SW(SR), or equivalent.**
P7	Connector	**MS3126F-10-6P, or equivalent**
R1	Resistor, variable	**5000 ohms ±10%, 2 w.**
R2	Resistor, fixed	**18 K** *Ohms* ±10%, 1/2 W.
Sl, S3	Switch, toggle	SPST
S2	Switch, toggle	4PST
TP1, TP2	Jack	H. H. Smith 1501, **or** equivalent.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph 11-2.1 apply. The receiver-transmitter shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the receiver-transmitter shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the receiver-transmitter shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be performed as specified in procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4*7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating) This test shall be performed as specified in Procedures I and II, "Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.4.2 Low **(operating and non-operating)** This test shall be performed as specified in Procedures I and II,Mthod 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810. The **operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.**

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4. Each conductor shall be electrically continuous as specified on the schematic diagram.

4.7.3 Performance. The selected receiver-transmitters shall be tested as follows:

4.7.3.1 Interlock switch. Using standard measuring equipment, verify that interlock fulfills the continuity requirements of 3.5.1.

4.7.3.2 Clutter lock-on switch. Using standard measuring equipment, verify that the clutter lock-on switch fulfills the continuity requirements of 3.5.2.

4.7.3.3 Overload reset switch. Using standard measuring equipment, verify that the overload reset switch fulfills the continuity requirements of 3.5.3.

4.7.3.4 Indicator lamps.

4.7.3.4.1 Modulator overload indicator. Using the test setup shown in Figure 2, verify that the modulator overload indicator meets the requirements of 3.5.4.1. .

4.7.3.4.2 High voltage overload indicator. Using the test setup shown in Figure 2, verify that the high voltage overload indicator meets the requirements of 3.5.4.2.

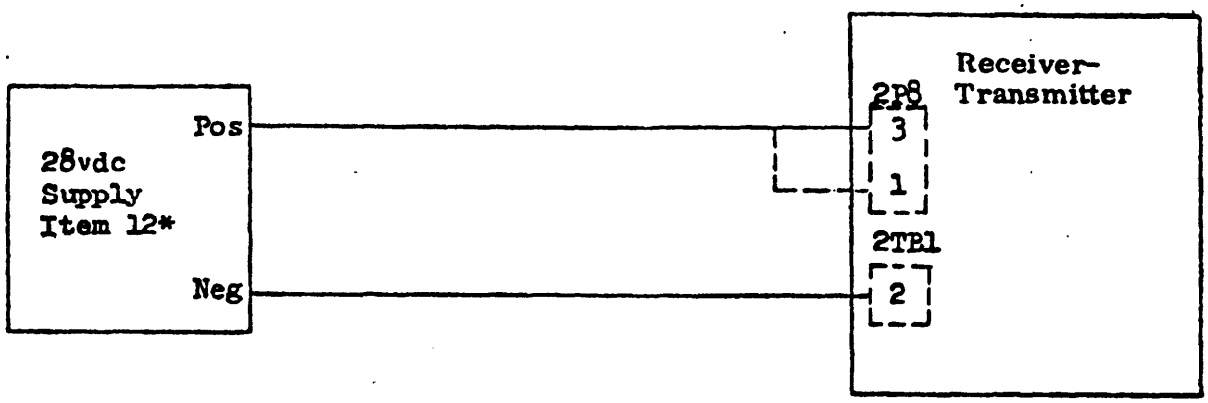
4.7.3.4.3 **Power** indicator lamp.

a. Connect the **receiver-transmitter in the test setup shown in Figure 3. Verify that all power supplies and all test fixture switches are turned to 0ff.**

b. Turn on power supply (item 13, Figure 3) and adjust for 24 V output.

*c** Place **the** interlock switch in the manual override position. Operate test fixture switch S1 to the on position. The power indicator lamp shall illuminate.

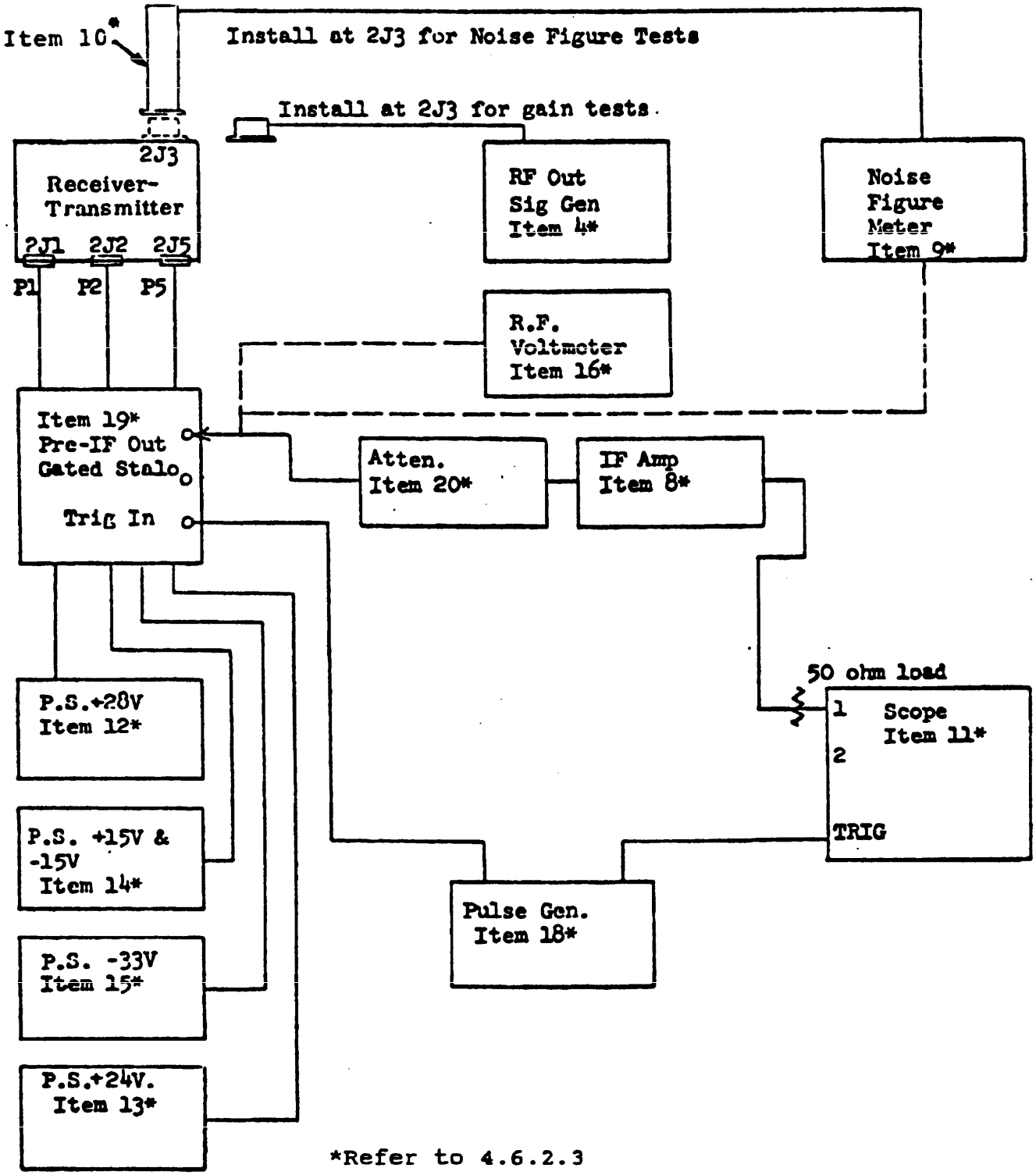
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*Refer to 4.6.2.3

FIGURE 2. Test setup.

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*Refer to 4.6.2.3

FIGURE 3. Test setup for standby operation.

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4.7.3.5 Standby operation. Adjust test equipment, shown in Figure 3, as required in Table V.

4.7.3.5.1 Blower operation. With the equipment operated as in 4.7.3.4.3, the blower shall operate and provide an intake airflow.

4.7.3.5.2 **Gates stalo output**.

4.7.3.5.2.1 Turn on all power supplies and connect a 50-ohm coaxial test cable between **gated stalo J3 output on test fixture and channel 2 input on the oscilloscope (item 11, Figure 3). Terminate the cable with a 50-ohm load at the oscilloscope. Turn on test fixture switch S2.**

4.7.3.5.2.2 Using the delayed sweep feature and external trigger from the pulse generator (item 18, Figure 3), adjust the oscilloscope for a 1 us time base. observe that the gated stalo output meets the requirements of 3.5.5.2. Disconnect the test cable.

4.7.3.5.3 pulse shaper output.

4.7.3.5.3.1 Disconnect 2P23 from 2A1J2 and connect a coaxial test cable between 2P23 and channel 2 input on the oscilloscope (item 11, Figure 3).

4.7.3.5.3.2 Verify that the pulse shaper output pulse at 2P23 meets the requirements of 3.5.5.3. Disconnect the test cable.

4.7.3.5.4 Overall gain.

4.7.3.5.4.1 Connect the signal generator (Item **4,** Figure 3) output to 2J3 on the receiver-transmitter and set the output to -40 dBm.

4.7.3.5.4.2 Disconnect 2P19 from 2AlA5Jl and connect the rf voltmeter (item 16, Figure 3) to 2A1A5J1.

4.7.3.5.4.3 Plug the channel 4 crystal module into the microwave assembly and tune the signal generator for a peak indication on the rf voltmeter. Set the AGC voltage control, R1 on the test fixture, for 0.0 V AGC. The rf voltmeter shall indicate not less than -23 dBm and not more than -18 dBm.

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TABLE V. Test equipment adjustments

TEST EQUIPMENT	ADJUST FOR
Pulse generator, item 18.	20 KHz PRF, pulse width 0.5 us, pulse amplitude +15 V peak.
Power supply, item 12.	+28 Vdc.
Power supply, item 14.	Output no. 1 for +15 Vdc. Output no. 2 for -15 Vdc.
Power supply, item 15.	-33 Vdc.
Power supply, item 13.	+24 Vdc.
other items.	As required by test procedure.

4.7.3.5.5 AGC action.

a. Disconnect the rf voltmeter, (item 16, Figure 3) and connect 2A1A5J1 to the attenuator (item 20, Figure 3), i-f amplifier (item 8, Figure 3) and oscilloscope (Item 11, Figure 3).

b. Set 20 dB attenuation in the attenuator and establish a reference level on the oscilloscope, peaking up the x-band signal generator tuning if necessary.

c. Advance the AGC voltage, using AGC voltage control, R1 on the test fixture, from 0.0 to +6.0 V at the same time reducing the attenuator to maintain reference level established in 4.7.3.5.5.2. At 6.0 V AGC and O dB attenuation, the signal output shall not exceed the reference level.

4.7.3.5.6 Noise figure.

a. Connect the x-band noise source waveguide (item 10, Figure 3) to 2J3 and connect the noise figure meter (item 9, Figure 3) to the source.

b. Connect a 50-ohm coaxial test cable from 2A1A5J1 to the i-f input on the noise figure meter.

c. Measure noise figure using the procedure given in the manufacturer's manual for the noise figure meter. Reading shall be not more than 12 dB.

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d. Repeat 4.7.3.5.6.3 installing each of the other five crystal modules, in turn, in the microwave assembly. The reading on each channel shall be not more than 12 dB. Disconnect the noise figure test equipment and reconnect 2P19 at 2A1A5J1.

4.7 3.5.7 Switch drive pulse. Connect a test cable to 2A1A7J1 and 2A1P5, using a tee connector, and connect the other end of the cable to the oscilloscope channel 2 (item 11, Figure 3). The switch drive pulse parameters shall meet the requirements of 3.5.5.6 ! Remove the test cable and tee connector and reconnect 2A1P5 to 2A1A7J1.

4.7.3.5.8 Diode switch isolation.

a. Connect the x-band signal generator (item 4, Figure 3) to 2J3. Connect a test cable from pre-amp i-f output J1 connector on the test fixture to channel 2 input on the oscilloscope (item 11, Figure 3) with a 50-ohm load on the oscilloscope end.

b. Set the x-band generator output level to -20 dBm and peak the frequency for maximum deflection on the oscilloscope.

c. Adjust the oscilloscope delayed sweep for a 5 us time base synchronous with pulse generator trigger (item 18, Figure 3).

d. With the display as shown in Figure 4, verify that the rf level-during the gated off time is 45 dB or greater than the level during the on time.

4.7.3.6 Radiate mode.

4.7.3.6.1 Average output power.

a. Connect the receiver-transmitter in the test setup as shown in Figure 5.

b. Turn on switch S3 of the test fixture and tune the klystron for maximum output at the operating frequency. The power meter (item 7, Figure 5) shall indicate not less than 0.33 mv. (Note: with 40 dB coupler, 0.426 mW at the 40 dB port indicates 4.26 W at through port.)

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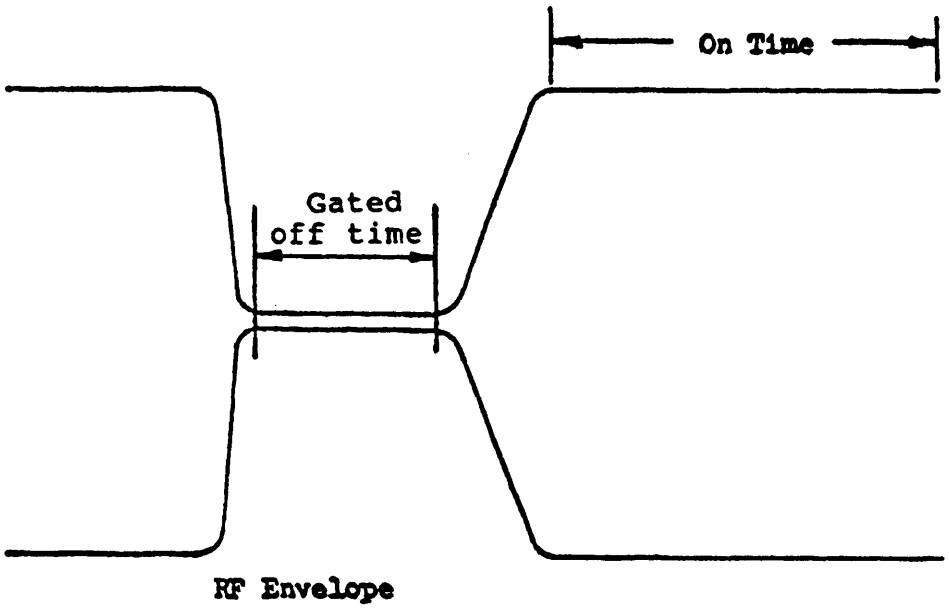


FIGURE 4. Diode Switch Isolation Pattern.

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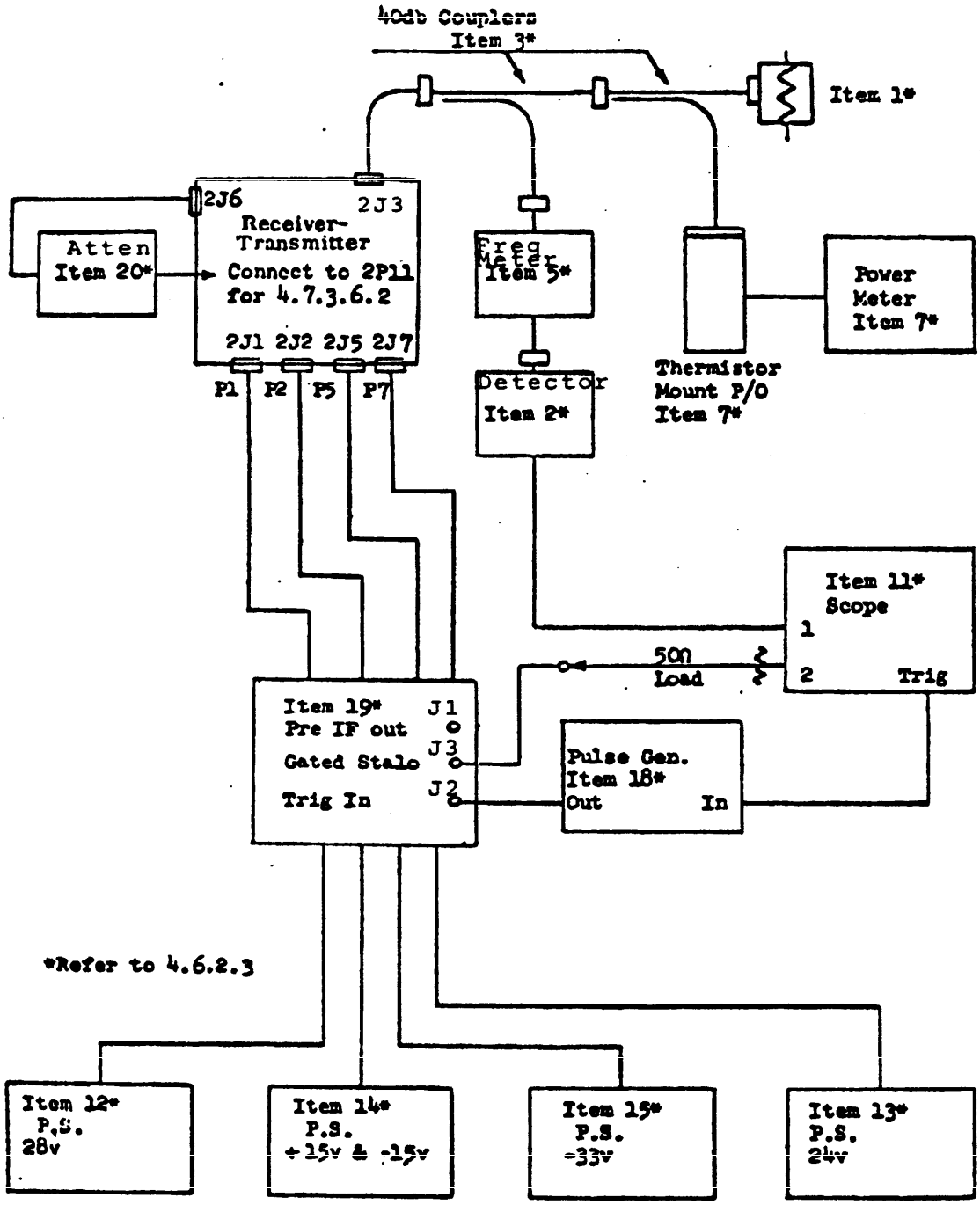


FIGURE 5. Test setup for radiate mode.

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c. Repeat 4.7.3.6.1.2 on each of the five channels. Turn off test fixture switch S3.

4.7.3.6.2 Transmit pulse timing.

a. Connect a coaxial test cable between 2J6 and the rf attenuator (item 20, Figure 5). Connect the output of the attenuator to 2P11 in the receiver-transmitter unit. Set the attenuator for 14 dB attenuation.

b. Turn on test fixture switch S3. With the oscilloscope (item 11, Figure 5) channel selector on "alternate" and a 1 us time base, adjust the delayed sweep to view the detected rf pulse and the gated stalo burst. The timing requirements of 3.5.6.2 shall be met. Turn off test fixture S3 and disconnect the attenuator from 2J6. Restore 2P11 to its original connection.

4.7.3.6.3 Transmit pulse envelope.

a. Turn on test fixture switch S3. Observe that the detected rf envelope on charnnel 1 of the oscilloscope meets the requirements of 3.5.6.3. Turn off switch S3.

4.7.3.6.4 Transmit frequency.

a. Turn on 2S6, service radiate switch. While observing the transmit pulse display, tune the cavity frequency meter (itam 5, figure 5) for a slight dip in pulse amplitude. At the point of maximum dip, read the frequency from the scale of the frequency meter. The reading shall meet the requirements of 3.5.6.4.

b. Repeat 4.7.3.6.4.1 with each of the other five crystal modules installed, tuning the klystron to each operating frequency.

4.7.3.6.5 Transmit indicator threshold.

a. Turn on the test fixture switch S3. Using the power meter (Item 7, Figure 5) measure and note the average transmitter power output at 2J3. Verify that the test fixture transmit indicator DS3 is illuminated.

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b. Remove the cap of attenuator 2A1AT1 and loosen setscrews. Adjust attenutor 2A1AT1 until test fixture transmit indicator DS3 extinguishes. Then readjust attenuator 2A1AT1 until test fixture transmit indictor DS3 just illuminates. The average transmitter power output at 2J3 as measured by (Item 7, Figure 5) shall be 1.8 ± 0.1 watts (W).

c. Readjust attenuator 2A1AT1 until transmitter power output is at the original level noted in 4.7.3.6.5.1. Lock attenuator setscrews and replace protective cap. Turn off test fixture switch S3.

4.7.3.7 Subclutter visibility (scv) test. After performing the test of 4.7.3 through 4.7.3.6.5.3, substitute the receiver-transmitter into a Radar Set AN/VPS-2 and perform the subclutter visibility (SCV) test as specified in 3.5.7. The Radar Set AN/VPS-2 and test facilities used shall be as specified in MIL-R-50651.

4.8 Soldering. Soldering **processes** shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 **as** specified in paragraph 3.7.

5* PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet PSI0554525. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Receiver-Transmitter, Radar, RT-860/VPS-2 is part of Radar Set AN/VPS-2. The receiver transmitter provides sufficient pulsed rf power to obtain usable echo returns from targets of interest and converts these target signals to an intermediate frequency for use in other units of the system.

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6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and **date of this specification.**
b. **Applicable stock number.**
c. **First Article Sample requirements.**
d. **Packaging requirements, if other than specified in Section 5.**
e. **Serialization requirements, if applicable.**
f* **Certificate of conformance for each lot of shopment of product.**

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-1, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract **Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.**

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. **Army** *Armament Research, Development and* Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Receiver-transmitter
Radar
Pulsed rf power

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A711)

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	MIL-R-50668A (AR)	14 September 1990

3. DOCUMENT TITLE
RECEUVER-TRANSMITTER, RADAR RT-860/VPS-2: 10548089

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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